Lctr 10: Convergence, Properties of CT Fourier Series, and DT Fourier Series

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Focus: Sections 3.4 – 3.7 (Pages 195 to 227)

10.1 Does the Fourier Series Always Work? Convergence

Why This Matters

In Lecture 9 we learned that periodic signals can be represented as sums of harmonically related complex exponentials (the Fourier series). But can **every** periodic signal be represented this way? And when we add up infinitely many terms, does the sum actually converge to the original signal? Understanding convergence tells us when and how we can trust the Fourier series representation—which is essential for using it in real engineering applications.

10.1.1 The Good News: Dirichlet Conditions

For essentially every periodic signal we encounter in engineering practice, the Fourier series representation is valid. The mathematical guarantee comes from the **Dirichlet conditions**. A periodic signal $x(t)$ with period T has a convergent Fourier series if it satisfies all three of the following conditions:

1. **Absolute integrability:** The signal has finite area over one period:

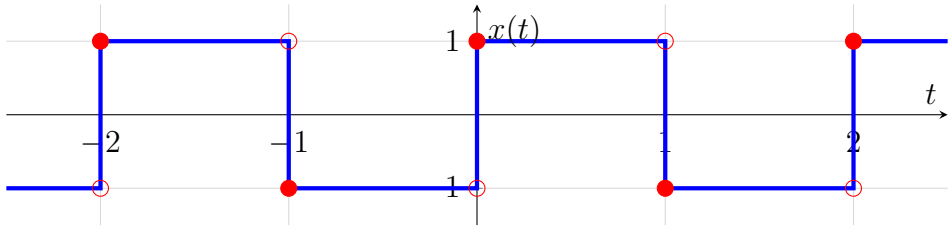
$$\int_T |x(t)|\, dt < \infty$$

   This ensures every coefficient a_k is finite.

2. **Bounded variation:** In any single period, $x(t)$ has a finite number of maxima and minima (no infinitely fast oscillations).

3. **Finite discontinuities:** In any single period, there are only a finite number of discontinuities, and each discontinuity has a finite jump.

Satisfies Dirichlet Conditions: Square Wave (finite jumps, finite area)



Violates Dirichlet Conditions: $x(t) = \sin(1/t)$ near $t = 0$ (infinite oscillations)

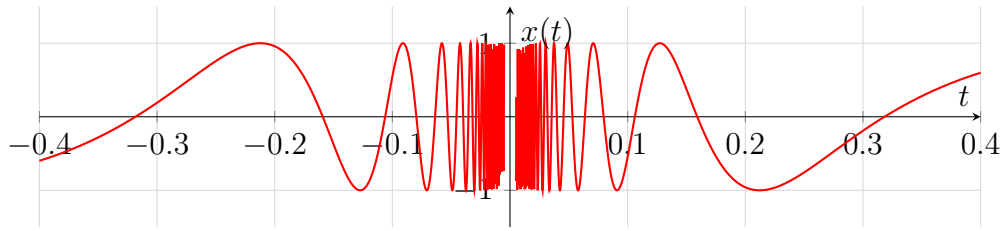


Figure 1: Top: A square wave satisfies all three Dirichlet conditions—it has finite area, a finite number of maxima/minima, and finite jumps. Bottom: $\sin(1/t)$ oscillates infinitely fast near $t = 0$, violating condition 2.

Key Insight

Practical takeaway: Every periodic signal you will encounter in this course—square waves, sawtooth waves, pulse trains, sinusoids, and so on—satisfies the Dirichlet conditions. The signals that violate these conditions are mathematically pathological and essentially never appear in real engineering systems.

10.1.2 What Happens at Discontinuities? The Gibbs Phenomenon

When a periodic signal has jump discontinuities (like a square wave), the Fourier series converges to the **midpoint** of the jump at the discontinuity. For example, if a square wave jumps from 0 to 1, the Fourier series converges to 0.5 at the discontinuity.

Furthermore, if we use only a **finite** number of harmonics (a truncated Fourier series), we observe **ripples** near the discontinuities. This is the **Gibbs phenomenon**:

- As we add more harmonics, the ripples get narrower and compress toward the discontinuity.
- However, the **peak amplitude** of the ripple does *not* decrease—it remains at about 9% of the jump height, no matter how many terms we include.
- In the limit of infinitely many terms, the ripples become infinitely narrow, and the energy in the error goes to zero.

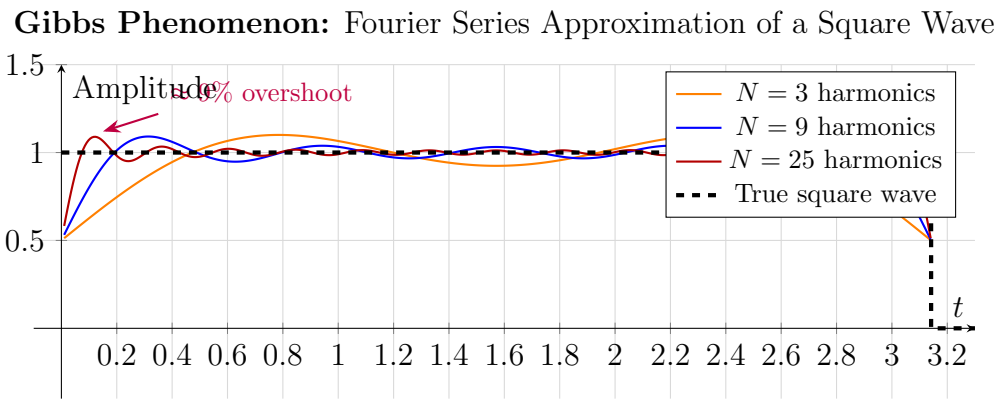


Figure 2: The Gibbs phenomenon in action. As we add more harmonics (orange → blue → red), the ripples become narrower and move closer to the discontinuity, but the peak overshoot remains ≈ 9%. The dashed line shows the true square wave.

Step-by-Step: Understanding the Gibbs Phenomenon

What is happening in the graph above?

1. **Start with few harmonics (orange, $N = 3$):** We have only the fundamental and a couple of odd harmonics. The approximation is very "wavy"—the sinusoids cannot make a sharp corner, so there are large oscillations and a noticeable overshoot right at the edge of the jump.

2. **Add more harmonics (blue, $N = 9$):** The approximation becomes sharper. The ripples are now smaller in width and more concentrated near the discontinuity. But the peak overshoot is still about the same height ($\approx 9\%$ above the true value).

3. **Many harmonics (red, $N = 25$):** The approximation looks quite close to the true square wave. The ripples are now very narrow spikes hugging the discontinuity. But zoom in on the jump—the peak is *still* $\approx 9\%$ above the line.

4. **The pattern:** No matter how many terms we add, the overshoot height stays at $\approx 9\%$. Only in the limit $N \to \infty$ does the ripple become infinitely narrow (zero width), so its energy goes to zero.

Common Mistake

The Gibbs phenomenon means that a finite Fourier series approximation will **always** overshoot at discontinuities by about 9%. This is not a bug—it is a fundamental property of how sinusoids approximate sharp jumps. In practical applications (like audio processing), this can sometimes matter and needs to be accounted for.

10.2 Properties of Continuous-Time Fourier Series (CT FS)

The Fourier series satisfies many useful properties. These let us find coefficients of complex signals by relating them to simpler signals whose coefficients we already know, *without* recomputing the analysis integral from scratch.

Throughout this section, we use the shorthand notation:

$$x(t)\text{FS}a_k$$

to indicate that $x(t)$ is a periodic signal with Fourier series coefficients a_k.

10.2.1 Linearity

If $x(t)\text{FS}a_k$ and $y(t)\text{FS}b_k$ (both with the same period T), then:

$$\boxed{Ax(t) + By(t)\text{FS}Aa_k + Bb_k}$$

Meaning: The Fourier coefficients of a linear combination are the same linear combination of the individual Fourier coefficients.

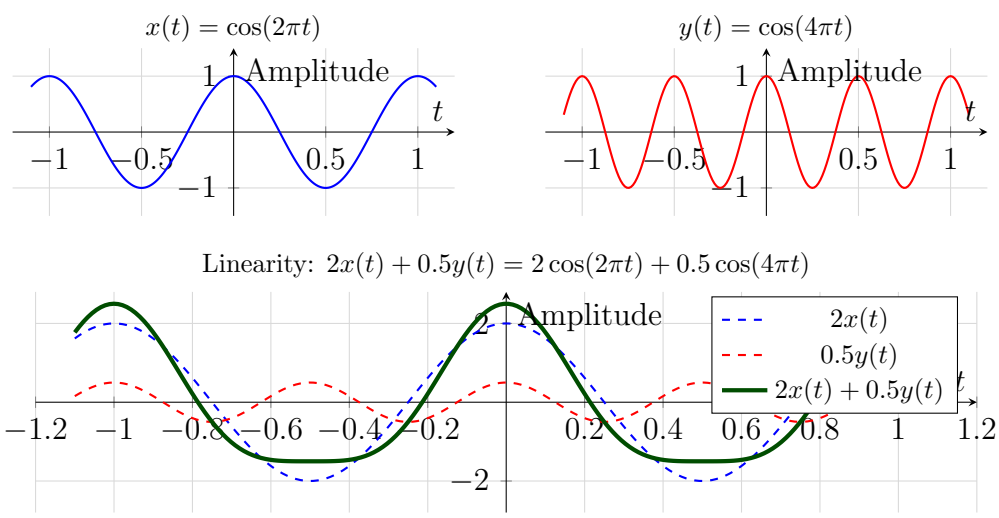


Figure 3: Linearity in action. Top: two individual cosines. Bottom: scaling and adding in the time domain corresponds to scaling and adding the Fourier coefficients: $Aa_k + Bb_k$.

10.2.2 Time Shifting

If $x(t)\text{FS}a_k$, then:

$$\boxed{x(t - t_0)\text{FS}a_k\, e^{-jk\omega_0 t_0}}$$

Meaning: Shifting a signal in time multiplies each Fourier coefficient by a complex exponential phase factor $e^{-jk\omega_0 t_0}$. Crucially, the **magnitude** $|a_k|$ does not change—only the

phase changes. Time shifting does not affect the "amount" of each harmonic, only *when* each harmonic starts.

Time Shift: Same shape, different starting time

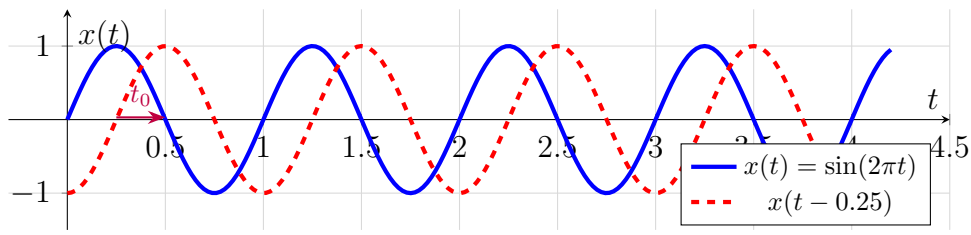


Figure 4: A time shift by $t_0 = 0.25$ slides the waveform to the right. The Fourier coefficient magnitudes $|a_k|$ are unchanged—only the phases rotate by $e^{-jk\omega_0 t_0}$.

Step-by-Step: Why Does Time Shift Only Change Phase?

1. Recall the synthesis equation: $x(t) = \sum_k a_k\, e^{jk\omega_0 t}$.
2. A time shift gives: $x(t - t_0) = \sum_k a_k\, e^{jk\omega_0 (t - t_0)} = \sum_k \underbrace{a_k\, e^{-jk\omega_0 t_0}}_{\text{new coefficient}}\, e^{jk\omega_0 t}$.
3. The new coefficient is $a_k\, e^{-jk\omega_0 t_0}$. Since $|e^{-jk\omega_0 t_0}| = 1$, we get $|a_k\, e^{-jk\omega_0 t_0}| = |a_k|$.
4. **Physical meaning:** Each harmonic still has the same "loudness" (magnitude), but each starts at a different phase—higher harmonics are shifted proportionally more.

10.2.3 Time Reversal

If $x(t)\text{FS}a_k$, then:

$$\boxed{x(-t)\text{FS}a_{-k}}$$

Meaning: Reversing a signal in time reverses the order of the Fourier coefficients.

Consequence: If $x(t)$ is **even** ($x(-t) = x(t)$), then $a_{-k} = a_k$, so the coefficients are also even. If $x(t)$ is **odd** ($x(-t) = -x(t)$), then $a_{-k} = -a_k$, so the coefficients are odd.

Time Reversal: $x(t)$ vs $x(-t)$

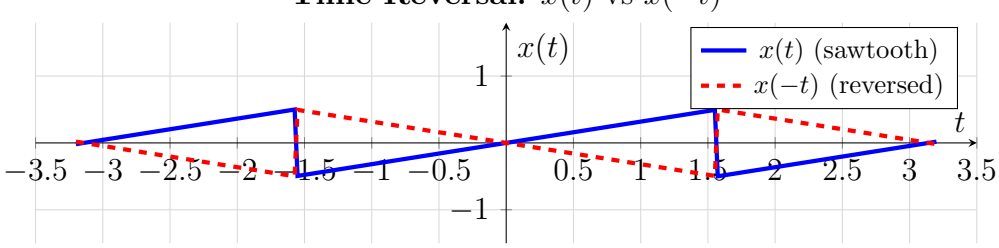


Figure 5: A sawtooth wave $x(t)$ and its time-reversed version $x(-t)$. The coefficients transform as $a_k \to a_{-k}$, i.e. the index reverses.

10.2.4 Differentiation

If $x(t)\text{FS}a_k$, then:

$$\boxed{\frac{dx(t)}{dt}\text{FS}jk\omega_0\, a_k}$$

Meaning: Differentiation in time corresponds to multiplying the kth coefficient by $jk\omega_0$. This amplifies higher harmonics (large $|k|$) relative to lower ones.

Practical use: If it is easier to find the Fourier coefficients of dx/dt than of $x(t)$ itself, compute those first and then divide by $jk\omega_0$ to recover a_k (for $k \neq 0$). Compute a_0 separately from the average value formula.

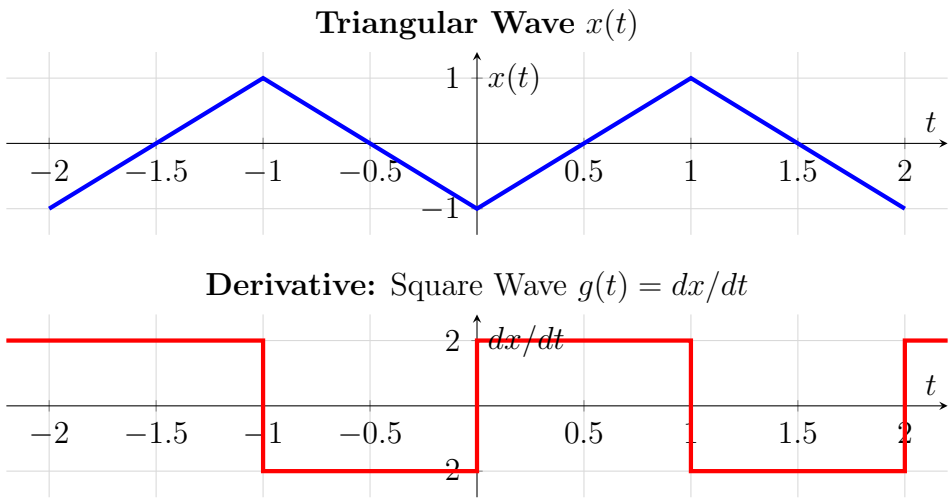


Figure 6: The differentiation property in visual form. The derivative of a triangular wave is a square wave. If we know the Fourier coefficients d_k of the square wave, we can recover the triangular wave coefficients as $a_k = d_k/(jk\omega_0)$ for $k \neq 0$.

10.2.5 Parseval's Relation

$$\boxed{\frac{1}{T}\int_T |x(t)|^2\, dt = \sum_{k=-\infty}^{+\infty} |a_k|^2}$$

Meaning: The average power of the signal over one period equals the sum of the powers in all its harmonic components. This is a statement of **energy conservation** in the frequency domain.

$|a_k|^2$ represents the average power in the kth harmonic. Parseval's relation says: the total power is distributed across the harmonics, and none is lost or created.

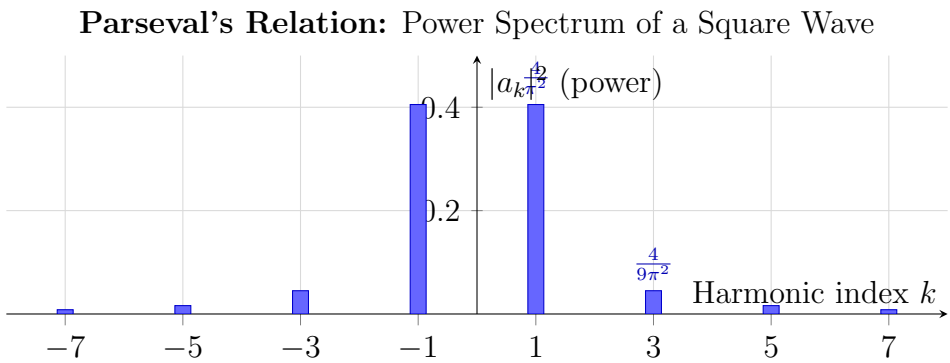


Figure 7: Power spectrum of a square wave. Each bar shows $|a_k|^2$, the power in the kth harmonic. Parseval's relation guarantees that the sum of all bars equals the average power computed directly from $x(t)$.

Step-by-Step: Parseval's Relation for a Square Wave

Consider a square wave oscillating between -1 and $+1$ with period T.

1. **Left side (time domain):** The signal is always $|x(t)| = 1$, so $|x(t)|^2 = 1$. The average power is:
$$\frac{1}{T}\int_T |x(t)|^2\,dt = \frac{1}{T}\int_T 1\,dt = 1$$

2. **Right side (frequency domain):** The Fourier coefficients are $a_k = \frac{2}{jk\pi}$ for odd k, $a_k = 0$ for even $k \neq 0$, and $a_0 = 0$. So:
$$\sum_{k=-\infty}^{+\infty} |a_k|^2 = \sum_{\substack{k \text{ odd} \\ k\neq 0}} \frac{4}{k^2\pi^2} = \frac{4}{\pi^2}\left(1 + \frac{1}{9} + \frac{1}{25} + \cdots\right) \times 2 = 1 \checkmark$$

3. **Conclusion:** Total power in time domain = total power across all harmonics. Energy is conserved—just redistributed across frequencies!

10.3 Worked Example Using Properties

10.3.1 Example: Fourier Coefficients via Differentiation

Consider the **triangular wave** $x(t)$ with period $T = 4$ and fundamental frequency $\omega_0 = \pi/2$, which ramps linearly up and down between -1 and $+1$.

Observation: The derivative of a triangular wave is a **square wave** $g(t)$ (alternating between $+1$ and -1).

Step-by-Step: Finding Triangular Wave Coefficients via Differentiation

Strategy: Instead of integrating $x(t)$ directly, differentiate to get a simpler signal, find *its* coefficients, then work backwards.

1. **Differentiate:** $g(t) = \dfrac{dx}{dt}$ is a square wave alternating between $+1$ (upslope) and -1 (downslope), with the same period $T = 4$.

2. **Find coefficients of $g(t)$:** From our known square wave results (Lecture 9), the Fourier coefficients d_k of this square wave are:

$$d_k = \begin{cases} \dfrac{2}{jk\pi} & k \text{ odd} \\ 0 & k \text{ even}, \ k \neq 0 \end{cases}$$

3. **Apply the differentiation property in reverse:** Since $g(t) = dx/dt$, we have $d_k = jk\omega_0 \cdot a_k$, where a_k are the coefficients of $x(t)$. Solving:

$$a_k = \frac{d_k}{jk\omega_0} = \frac{d_k}{jk(\pi/2)} = \frac{2\,d_k}{jk\pi}, \quad k \neq 0$$

4. **Substitute d_k:** For odd k:

$$a_k = \frac{2}{jk\pi} \cdot \frac{2}{jk\pi} = \frac{4}{(jk\pi)^2} = \frac{4}{-k^2\pi^2} = -\frac{4}{k^2\pi^2}$$

   For even $k \neq 0$: $a_k = 0$.

5. **Compute a_0 separately:** The average value of the triangular wave is $a_0 = 0$ (symmetric about zero).

6. **Final result:**

$$\boxed{a_k = \begin{cases} -\dfrac{4}{k^2\pi^2}, & k \text{ odd} \\ 0, & k \text{ even} \end{cases}}$$

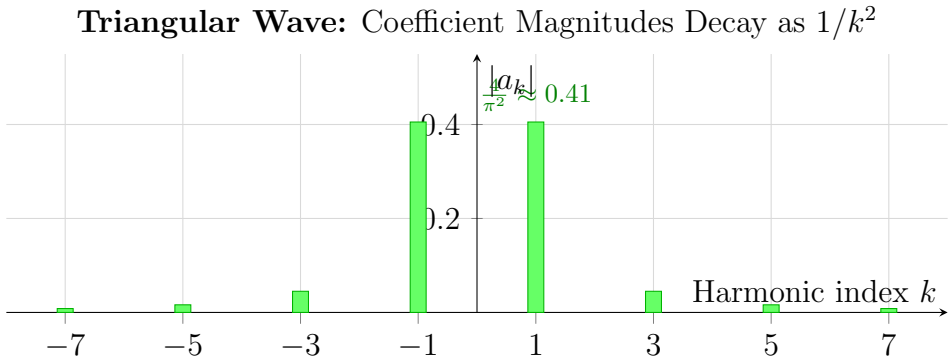


Figure 8: Triangular wave coefficients decay as $1/k^2$, much faster than the square wave's $1/k$ decay. This is because the triangular wave is smoother (continuous), and smoother signals have faster-decaying Fourier coefficients.

Key Insight

Rule of thumb: The smoother a signal, the faster its Fourier coefficients decay. Discontinuous signals (like square waves) have coefficients that decay as $1/k$. Continuous signals with discontinuous derivatives (like triangular waves) decay as $1/k^2$. Smoother signals decay even faster. This is why smooth signals need fewer harmonics for a good approximation.

This demonstrates the power of using Fourier series properties: instead of integrating directly, we relate the coefficients of a complicated signal to those of a simpler one.

10.4 The Discrete-Time Fourier Series (DT FS)

Why This Matters

We now extend Fourier series to **discrete-time** periodic signals. The ideas are very similar to the continuous-time case, but with one major simplification: the discrete-time Fourier series is a **finite** sum, not an infinite one. This means there are **no convergence issues** in discrete time!

10.4.1 Periodicity in Discrete Time

A discrete-time signal $x[n]$ is periodic with period N if:

$$x[n] = x[n+N] \quad \text{for all } n$$

The fundamental period is the smallest positive integer N for which this holds, and the fundamental frequency is $\omega_0 = 2\pi/N$.

10.4.2 Harmonically Related Discrete-Time Complex Exponentials

The set of harmonically related complex exponentials in discrete time is:

$$\phi_k[n] = e^{jk(2\pi/N)n}, \quad k = 0, \pm 1, \pm 2, \ldots$$

Critical difference from continuous time: In discrete time, complex exponentials that differ in frequency by a multiple of 2π are **identical**:

$$e^{j(k+N)(2\pi/N)n} = e^{jk(2\pi/N)n} \cdot \underbrace{e^{j2\pi n}}_{=1 \text{ for all integer } n} = e^{jk(2\pi/N)n}$$

This means there are only N **distinct** harmonics: $\phi_0[n], \phi_1[n], \ldots, \phi_{N-1}[n]$. Adding more does not create new signals—they just repeat.

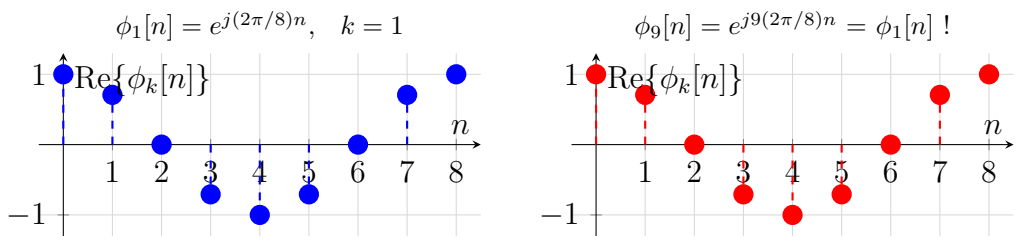


Figure 9: With $N = 8$: $\phi_1[n]$ and $\phi_9[n]$ produce *identical* sample values. Since $9 = 1 + 8$, and $e^{j2\pi n} = 1$ for integer n, the $k = 9$ harmonic is the same as $k = 1$. Only $N = 8$ distinct harmonics exist.

Key Insight

Key difference between CT and DT Fourier series:

- **CT (Continuous Time):** Infinitely many distinct harmonics → infinite series.
- **DT (Discrete Time):** Only N distinct harmonics → finite series with exactly N terms.

Because the DT Fourier series is a finite sum, it has **no convergence problems** and **no Gibbs phenomenon**.

10.4.3 The DT Fourier Series Pair

For a discrete-time periodic signal $x[n]$ with period N and fundamental frequency $\omega_0 = 2\pi/N$:

Synthesis equation:

$$\boxed{x[n] = \sum_{k=\langle N \rangle} a_k \, e^{jk(2\pi/N)n}}$$

Analysis equation:

$$\boxed{a_k = \frac{1}{N} \sum_{n=\langle N \rangle} x[n] \, e^{-jk(2\pi/N)n}}$$

where $\sum_{k=\langle N\rangle}$ and $\sum_{n=\langle N\rangle}$ mean "sum over any N consecutive values" (for example, $k = 0, 1, \ldots, N-1$, or $k = -2, -1, 0, 1, 2$ if $N = 5$).

Important: The DT Fourier coefficients a_k are themselves **periodic** with period N:

$$a_{k+N} = a_k$$

This makes sense because there are only N distinct harmonics, so only N distinct coefficients are needed.

10.4.4 Why Does the Analysis Equation Work?

The derivation mirrors the continuous-time case and relies on the discrete-time orthogonality relation:

$$\sum_{n=\langle N\rangle} e^{jk(2\pi/N)n} = \begin{cases} N, & \text{if } k = 0, \pm N, \pm 2N, \ldots \\ 0, & \text{otherwise} \end{cases}$$

This says: summing a discrete-time complex exponential over one complete period gives zero, *unless* the exponential is a constant (which happens when k is a multiple of N).

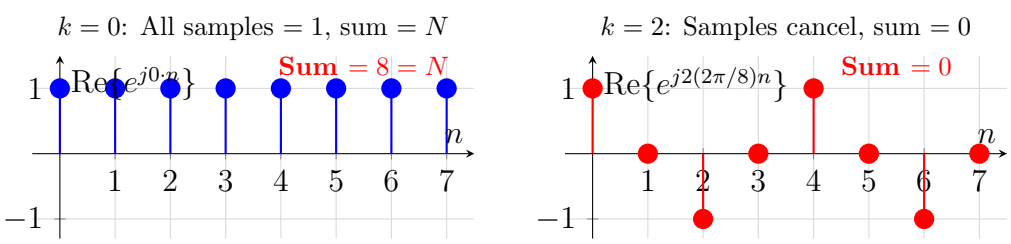


Figure 10: DT orthogonality with $N = 8$. Left: $k = 0$ gives all ones, summing to N. Right: $k = 2$ produces samples that cancel perfectly when summed, giving zero. This orthogonality is what makes the analysis equation work.

10.5 DT Fourier Series Examples

10.5.1 Example: $x[n] = \sin\left(\frac{2\pi}{N}n\right)$

This is the discrete-time counterpart of Example 9.5.1.

Step-by-Step: DT Sine Wave Coefficients

Goal: Find the Fourier coefficients of $x[n] = \sin\left(\frac{2\pi}{N}n\right)$.
Step 1: Expand using Euler's relation:

$$\sin\left(\frac{2\pi}{N}n\right) = \frac{1}{2j}\,e^{j(2\pi/N)n} - \frac{1}{2j}\,e^{-j(2\pi/N)n}$$

Step 2: Compare with the synthesis equation $x[n] = \sum_{k=\langle N\rangle} a_k\, e^{jk(2\pi/N)n}$:

- The first term has $k = 1$: $\quad a_1 = \dfrac{1}{2j}$
- The second term has $k = -1$: $\quad a_{-1} = -\dfrac{1}{2j}$
- All other $a_k = 0$ within one period.

Step 3 — Periodicity of coefficients: Since a_k is periodic with period N:

$$a_{N-1} = a_{-1} = -\frac{1}{2j}, \quad a_{N+1} = a_1 = \frac{1}{2j}, \quad \text{etc.}$$

Interpretation: A pure DT sinusoid has exactly two non-zero coefficients ($k = \pm 1$), just like in continuous time.

DT Sine Wave: Only two non-zero coefficients per period ($N = 8$ shown)

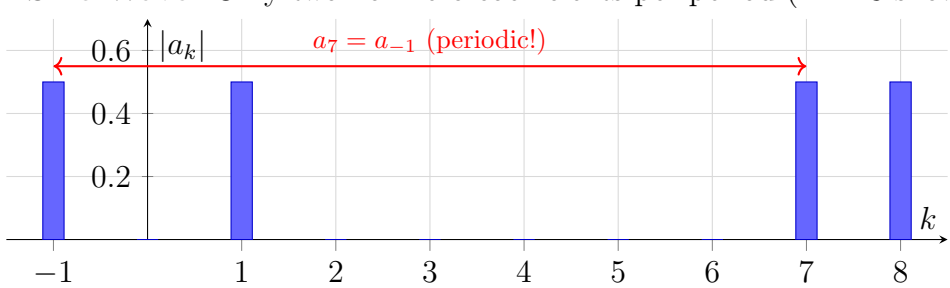


Figure 11: Magnitude spectrum of $\sin(2\pi n/8)$ with $N = 8$. Only $k = 1$ and $k = -1$ (equivalently $k = 7$) are non-zero, each with magnitude $1/2$.

10.5.2 Example: Discrete-Time Periodic Square Wave

Consider the DT periodic square wave: $x[n] = 1$ for $|n| \le N_1$ within one period of length N, and $x[n] = 0$ otherwise.

DT Periodic Square Wave: $N = 10$, $N_1 = 2$ (five "on" samples per period)

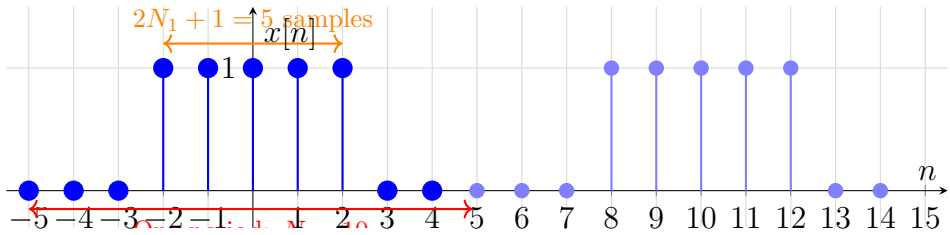


Figure 12: DT periodic square wave with $N = 10$ and $N_1 = 2$. Within each period, $2N_1 + 1 = 5$ samples are "on" (value 1) and the remaining 5 are "off" (value 0).

Using the analysis equation and evaluating the geometric series:

$$a_k = \begin{cases} \dfrac{1}{N} \cdot \dfrac{\sin\left[2\pi k(N_1 + \frac{1}{2})/N\right]}{\sin(\pi k/N)}, & k \neq 0, \pm N, \pm 2N, \ldots \\ \dfrac{2N_1 + 1}{N}, & k = 0, \pm N, \pm 2N, \ldots \end{cases}$$

DT Square Wave Coefficients: $N = 10$, $N_1 = 2$

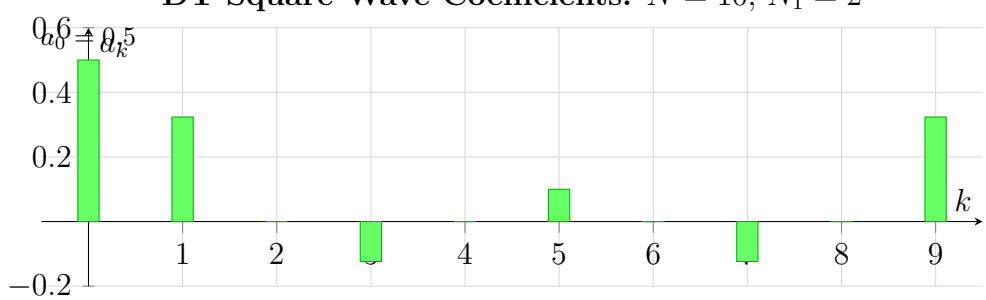


Figure 13: Fourier coefficients of the DT square wave. Note: only $N = 10$ coefficients per period are needed (they repeat with period N). Compare with the CT case, which requires infinitely many.

Key Insight

The DT square wave coefficients look similar to the CT square wave coefficients (both involve a $\sin/\sin$ or $\sin/k$ pattern). The key structural difference is that only N coefficients are needed per period in discrete time, while infinitely many are needed in continuous time.

10.6 Properties of Discrete-Time Fourier Series

The properties of the DT Fourier series closely mirror those of the CT Fourier series. Here are the most important ones.

Notation: $x[n]\text{FS}a_k$ means $x[n]$ has period N and Fourier coefficients a_k.

10.6.1 Linearity

$$Ax[n] + By[n] \text{FS} Aa_k + Bb_k$$

10.6.2 Time Shifting

$$x[n - n_0] \text{FS} a_k \, e^{-jk(2\pi/N)n_0}$$

10.6.3 First Difference

The DT analog of differentiation is the **first difference**: $x[n] - x[n-1]$.

$$x[n] - x[n-1] \text{FS} (1 - e^{-jk(2\pi/N)}) \, a_k$$

This is the discrete-time replacement for the CT differentiation property.

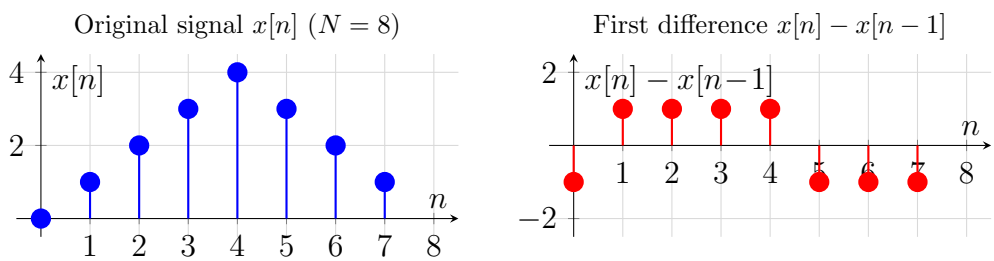


Figure 14: First difference in action. Left: a DT ramp-like signal. Right: the first difference $x[n]-x[n-1]$ acts like a discrete derivative, revealing the "slope" at each sample. Coefficients transform as $a_k \to (1 - e^{-jk(2\pi/N)})a_k$.

10.6.4 Parseval's Relation (DT)

$$\boxed{\frac{1}{N} \sum_{n=\langle N \rangle} |x[n]|^2 = \sum_{k=\langle N \rangle} |a_k|^2}$$

Just as in continuous time, this states that the average power per period equals the sum of the powers in all harmonic components. In DT, the sum on the right is finite (N terms).

10.6.5 Multiplication (Periodic Convolution of Coefficients)

If $x[n] \text{FS} a_k$ and $y[n] \text{FS} b_k$, both with period N, then:

$$x[n] \, y[n] \text{FS} d_k = \sum_{l=\langle N \rangle} a_l \, b_{k-l}$$

Meaning: Multiplying two signals in the time domain corresponds to **periodic convolution** of their Fourier coefficients. (This is similar to the multiplication property of CT Fourier series.)

10.7 Summary Table: CT vs. DT Fourier Series

	Continuous Time	**Discrete Time**
Period	T (any positive real number)	N (a positive integer)
Fund. frequency	$\omega_0 = 2\pi/T$	$\omega_0 = 2\pi/N$
Synthesis	$\displaystyle\sum_{k=-\infty}^{+\infty} a_k\, e^{jk\omega_0 t}$	$\displaystyle\sum_{k=\langle N\rangle} a_k\, e^{jk(2\pi/N)n}$
Analysis	$\displaystyle\frac{1}{T}\int_T x(t)\, e^{-jk\omega_0 t}\, dt$	$\displaystyle\frac{1}{N}\sum_{n=\langle N\rangle} x[n]\, e^{-jk(2\pi/N)n}$
Number of harmonics	Infinite	N (finite)
Convergence issues?	Yes (Dirichlet conditions, Gibbs phenomenon)	No
Coefficients periodic?	No	Yes, with period N

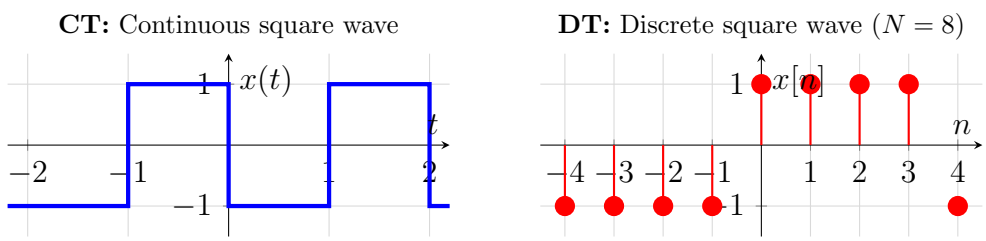


Figure 15: CT vs DT square waves side by side. The CT version is a continuous function of time (needing ∞ harmonics). The DT version is defined only at integer sample points (needing only N harmonics for a perfect representation).

10.8 Common Mistakes to Avoid

1. **Using infinite limits for DT Fourier series:** The DT synthesis sum has only N terms. Do not write $\sum_{k=-\infty}^{+\infty}$ for the DT case.

2. **Forgetting that DT coefficients are periodic:** $a_k = a_{k+N}$ always. So a_{-1} and a_{N-1} are the *same* coefficient.

3. **Mixing up CT and DT analysis formulas:** CT uses an *integral* divided by T. DT uses a *sum* divided by N. Don't swap them.

4. **Ignoring the Gibbs phenomenon:** When approximating a discontinuous CT signal with a finite number of harmonics, expect overshoot near discontinuities. Do not assume the approximation will be smooth.

5. **Forgetting a_0 when using the differentiation or first-difference property:** These properties give you a_k for $k \neq 0$ only. Always compute a_0 (the average value) separately.

Key Insight

Sanity check for Parseval's relation: Compute the average power of $x(t)$ or $x[n]$ directly (left side) and compare with $\sum |a_k|^2$ (right side). If they do not match, recheck your coefficients!

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